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(continued after index)

Per Kragh Andersen
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Statistical Models Based on Counting Processes

With 128 Illustrations

Springer

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Preface

One of the most remarkable examples of fast technology transfer from new developments in mathematical probability theory to applied statistical methodology is the use of counting processes, martingales in continuous time, and stochastic integration in *event history analysis*. By this (or generalized survival analysis), we understand the study of a collection of individuals, each moving among a finite (usually small) number of states. A basic example is moving from alive to dead, which forms the basis of *survival analysis*. Compared to other branches of statistics, this area is characterized by the dynamic temporal aspect, making modelling via the intensities useful, and by the special patterns of incompleteness of observation, of which *right-censoring* in survival analysis is the most important and best known example.

In this book our aim is to present a theory for handling such statistical problems, when observation is assumed to happen in continuous time. In most of the book, a minimum of assumptions are made (non- and semi-parametric theory), although one chapter considers parametric models. The detailed plan of the book is described in Chapter I.

Our desire has been to give an exposition in terms of the necessary mathematics (including previously unfamiliar tools that have proved useful), but at the same time illustrate most methods by concrete practical applications, primarily from our own biostatistical experience. We also wanted to indicate some recent lines of research. As a result, the book moves along at varying levels of mathematical sophistication and also at varying levels of subject-matter discussion. We realize that this will frustrate some readers, in particular those who had expected to find a smooth textbook. In Section I.1, we suggest a reading guide for various categories of readers, and in Section I.2, we give a synopsis of the developments of the field up to now.

The suggestion of writing a monograph on this material came from Klaus Krickeberg in 1982. In assembling a group of authors, we wanted to include our colleague and friend Odd O. Aalen, who started the theory and keeps contributing to it; however, Odd wanted to concentrate on building up medical statistics in Oslo, Norway. The work on the book has become something of a life-style for the four of us, involving drafting sections individually and then meeting, usually in Copenhagen, to discuss in great detail the merits and otherwise of the latest contributions. These meetings always lasted for three full days, with discussions going on at the dinner table to the wonder of our wives and children, none of whom believed we really wanted (were able to?) finish the project. We will miss these opportunities for very positive professional and human interaction, and the concept of "Vol. II" has become our consolation. (Do not worry, this will take a while!)

We have enjoyed extremely positive interest from all colleagues. First, our Nordic colleagues who participated in getting the theory developed have always been ready to follow up with good advice. Besides Odd Aalen, we have interacted particularly with Martin Jacobsen, as well as with Elja Arjas, Nils Lid Hjort, Søren Johansen, and Gert Nielsen. Overseas, we owe particular thanks to Dorota Dabrowska, who read several chapters in great detail and kept us informed of several fascinating new developments. We have been given good advice and been entrusted with prepublication versions of new results by many colleagues, including Chang-Jo F. Chung, Hani Doss, Priscilla Greenwood, Piet Groeneboom, Jean Jacod, John P. Klein, Tze Leung Lai, Ian McKeague, Jens Perch Nielsen, Michael Sørensen, Yangin Sun, Aad van der Vaart, Bert van Es, Mark van Pul, Wolfgang Wefelmeyer, and Jon A. Wellner.

As with any effort of this kind, countless drafts have been produced, some before the age of mathematical word processing. Although later phases have sent (most of) the authors themselves to the TeX keyboards (in Copenhagen with much support from Peter Dalgaard), many versions were done by our secretaries in Copenhagen (in particular, Susanne Kragskov and Kathe Jensen), Oslo (Dina Haraldsson and Inger Jansen), Amsterdam (Carolien Pol-Swagerman), and Utrecht (Karin Berlang and Diana van Doorn). Most calculations and graphs were done (in S) by Lars Sommer Hansen and Klaus Krøier, and Mark van Pul and Kristian Merckoll did some additional calculations.

Our work with the book has been done as part of our appointments at the University of Copenhagen (Per Kragh Andersen, Niels Keiding, the latter for part of the period as Danish Ministry of Education Research Professor), the University of Oslo (Ørnulf Borgan), and Centre for Mathematics and Computer Science, Amsterdam, and the University of Utrecht (Richard D. Gill). Some work was done during visits to the Department of Biostatistics, University of Washington; the Medical Research Council Biostatistics Unit, Cambridge, England; the Department of Mathematics, University of Western Australia; and the Department of Statistics, The Ohio State University.

We acknowledge the hospitality and interest everywhere. Travel costs and secretarial and computing assistance have been financed by our institutions, with supplements from the Danish Medical Research Council (grant no. 12-9229), the Norwegian Council for Research in Science and the Humanities, and Johan and Mimi Wessmann's foundation.

February 1992

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